

# Inference for instrumental variables

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Fall 2025

# Outline

- Asymptotics of reduced form parameters.
- Delta method for the Wald estimator.
- Weak instruments and failure of the delta method.
- Anderson-Rubin test and confidence sets

## Takeaways for this part of class

- Standard asymptotics give an approximate normal distribution for the Wald estimator.
- This is a bad approximation in the case of weak instruments: The remainder of a first-order expansion is too large.
- Valid inference is still possible, using likelihood ratio tests.

## Wald estimator

- Recall the 2SLS / Wald estimand:

$$\beta = \frac{\alpha}{\gamma},$$

where

$$\alpha = \text{Cov}(Z, Y), \quad \gamma = \text{Cov}(Z, D).$$

- Sample analog:

$$\hat{\beta} = \frac{\hat{\alpha}}{\hat{\gamma}},$$

where

$$\hat{\alpha} = \text{Cov}_n(Z, Y), \quad \hat{\gamma} = \text{Cov}_n(Z, D).$$

## Asymptotic distribution of the reduced form

For some variance matrix  $\Sigma$ ,

$$\sqrt{n} \cdot \begin{pmatrix} \hat{\alpha} - \alpha \\ \hat{\gamma} - \gamma \end{pmatrix} \rightarrow N(0, \Sigma)$$

### Practice problem

- Prove this, using the law of large numbers and the central limit theorem.
- You can assume  $E[Z] = 0$  and  $\hat{\alpha} = E_n[Z \cdot Y]$ ,  $\hat{\gamma} = E_n[Z \cdot D]$  for simplicity.

## Sketch of solution

- Let  $W_i = (Z_i Y_i, Z_i D_i)$ .
- The multivariate central limit gives

$$\frac{1}{\sqrt{n}} \cdot \sum_i (W_i - E[W_i]) \rightarrow N(0, \Sigma),$$

where  $\Sigma = \text{Var}(W_i)$ .

- The claim follows.
- For the general case ( $E[Z] \neq 0$ ), the delta-method yields the same result.

## Asymptotic distribution of the Wald estimator

Let again  $\hat{\beta} = \frac{\hat{\alpha}}{\hat{\gamma}}$ , where  $\hat{\alpha} = \text{Cov}_n(Z, Y)$ ,  $\hat{\gamma} = \text{Cov}_n(Z, D)$ .

Then, for some  $\sigma^2$ ,

$$\sqrt{n} \cdot (\hat{\beta} - \beta) \rightarrow N(0, \sigma^2).$$

### Practice problem

- Prove this, using
  - asymptotic normality of the reduced form,
  - and the delta method.
- What is  $\sigma^2$ ?

## Sketch of solution

- Let  $f(a, c) = \frac{a}{c}$ , so that  $\hat{\beta} = f(\hat{\alpha}, \hat{\gamma})$ .
- Consider the first-order Taylor approximation  $f(a, c) \approx f(\alpha, \gamma) + df \cdot \begin{pmatrix} a - \alpha \\ c - \gamma \end{pmatrix}$ ,  
where  $df = \left( \frac{\partial}{\partial a} f(\alpha, \gamma), \frac{\partial}{\partial c} f(\alpha, \gamma) \right) = \left( \frac{1}{\gamma}, -\frac{\alpha}{\gamma^2} \right)$ .
- Recall that  $\sqrt{n} \cdot \begin{pmatrix} \hat{\alpha} - \alpha \\ \hat{\gamma} - \gamma \end{pmatrix} \rightarrow N(0, \Sigma)$ .
- The delta method gives  $\sqrt{n} \cdot (f(\hat{\alpha}, \hat{\gamma}) - f(\alpha, \gamma)) \rightarrow N(0, \sigma^2)$ , where  $\sigma^2 = df \cdot \Sigma \cdot df'$ .



## Not so fast!

Suppose the asymptotic distribution of the reduced form is exact, so that

$$\begin{pmatrix} \hat{\alpha} - \alpha \\ \hat{\gamma} - \gamma \end{pmatrix} \sim N\left(0, \frac{1}{n}\Sigma\right)$$

### Practice problem

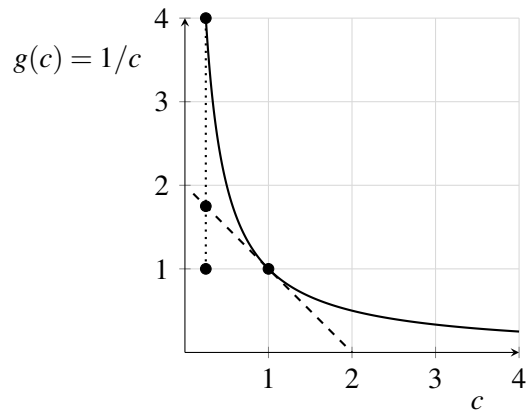
- Under this assumption, is  $\hat{\beta} = \frac{\hat{\alpha}}{\hat{\gamma}}$  unbiased for  $\beta$ ?
- Does the expectation even exist?
- Hint: Consider the special case of  $\alpha = \beta = 0$ , and look up the Cauchy distribution.

## When the delta-method fails

- The key step in our asymptotic normal approximation for  $\hat{\beta}$  was a first-order approximation to the function  $f(a, c) = \frac{a}{c}$ ,  
$$f(a, c) \approx f(\alpha, \gamma) + \frac{\partial}{\partial a} f(\alpha, \gamma) \cdot (a - \alpha) + \frac{\partial}{\partial c} f(\alpha, \gamma) \cdot (c - \gamma).$$
- First-order approximations are poor when the function is very non-linear.
- For intuition, consider a first-order approximation to the function  $g(c) = \frac{1}{c}$ ,  
$$g(c) \approx \tilde{g}(c) = \frac{1}{\gamma} - \frac{1}{\gamma^2} \cdot (c - \gamma).$$
- This approximation gets infinitely bad when  $c$  is close to 0:

$$\frac{g(c) - g(\gamma)}{\tilde{g}(c) - \tilde{g}(\gamma)} = \frac{\frac{1}{c} - \frac{1}{\gamma}}{-\frac{1}{\gamma^2} \cdot (c - \gamma)} = \frac{\frac{1}{\gamma c} \cdot (c - \gamma)}{\frac{1}{\gamma^2} \cdot (c - \gamma)} = \frac{\gamma}{c}.$$

## Illustration



## Testing null hypotheses on $\beta$

- Consider the null hypothesis  $\beta = \frac{\alpha}{\gamma} = \beta_0$ .
- Equivalently,  $\alpha - \gamma \cdot \beta_0 = 0$ .
- Under this null hypothesis,

$$\hat{\alpha} - \hat{\gamma} \cdot \beta_0 \sim N(0, \tau_{\beta_0}^2),$$

where

$$\tau_{\beta_0}^2 = \Sigma_{\alpha, \alpha} - 2\beta_0 \cdot \Sigma_{\alpha, \gamma} + \beta_0^2 \cdot \Sigma_{\gamma, \gamma}.$$

- This suggests a t-test which rejects if

$$\left| \frac{\hat{\alpha} - \hat{\gamma} \cdot \beta_0}{\hat{\tau}_{\beta_0}^2} \right| > z_{.975}.$$

- This is known as the *Anderson-Rubin* test.

## Confidence sets using test inversion

- Like any test, the Anderson-Rubin test can be inverted to give a confidence set for  $\beta$ :

$$CI = \left\{ \beta_0 : \left| \frac{\hat{\alpha} - \hat{\gamma} \cdot \beta_0}{\hat{\tau}_{\beta_0}^2} \right| > z_{.975} \right\}$$

- Note that  $CI$  is not necessarily a connected interval:
  - For  $\frac{\hat{\gamma}}{\sqrt{\Sigma_{\gamma,\gamma}}}$  close to 0,  $CI = (-\infty, \infty)$ .
  - For  $\frac{\hat{\gamma}}{\sqrt{\Sigma_{\gamma,\gamma}}}$  large,  $CI$  is approximately the delta-method based interval.
  - In intermediate cases,  $CI = (-\infty, a] \cup [b, \infty)$ .

# References

- *Andrews, I., Stock, J., and Sun, L. (2024). Weak instruments in IV regression: Theory and practice. Annual Review of Economics.*
- *Kasy, M. (2019). Uniformity and the delta method. Journal of Econometric Methods, 8(1).*